Analysis of Time Series Models for Electricity Consumption Forecasting

Qianqian Yao 1253177

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Abstract

To address climate change, we can discover catalysts to develop new energy resources, on the other hand we also should save the current energy resources. Predicting the electric power consumption precisely on demand can save energy. The electric power distribution problem is the distribution of electricity to different areas depends on its sequential usage, but predicting the following demand of a specific area is difficult, as it varies with weekdays, holidays, seasons, weather, temperatures, etc. It causes unnecessary waste of electricity and equipment depreciation. There is one efficient strategy to solve this problem is to predict the electrical transformers' oil temperature because the oil temperature can reflect the electricity transformer. The dataset is historical records of oil temperatures measured at electricity distribution stations. This project focuses on time series data analysis, forecasting model selection and evaluation including ARIMA, ARCH and GARCH models.

Keywords: Long-term time series, electricity consumption forecasting, ARIMA, ARCH, GARCH, Model Estimation